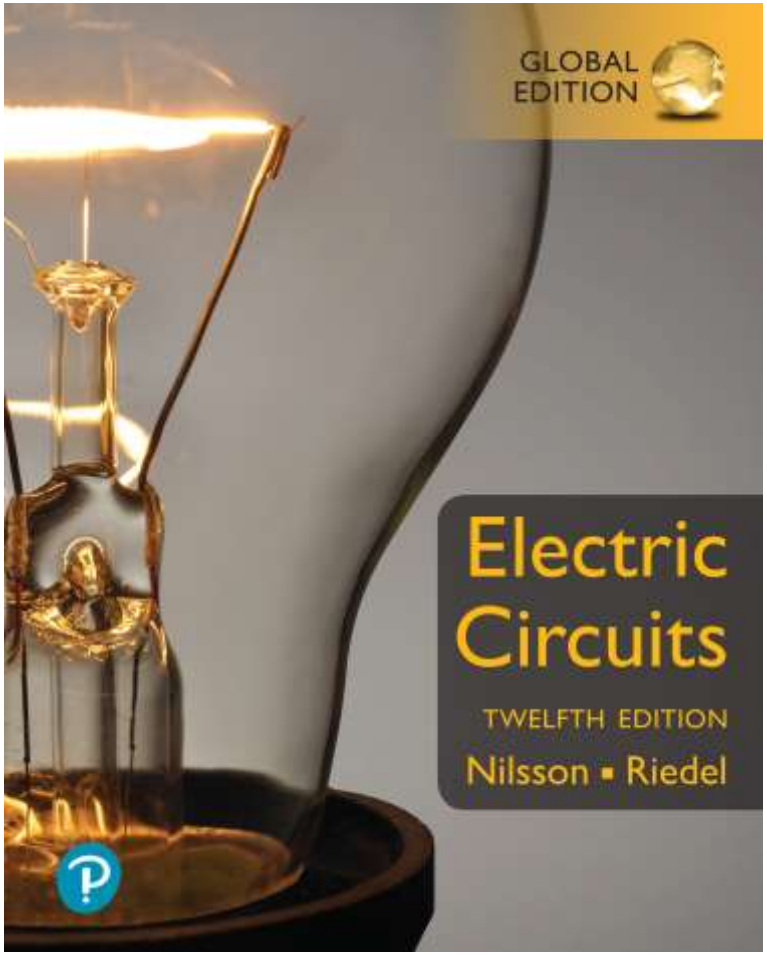
Electric Circuits

Twelfth Edition, Global Edition



Chapter 2 – Part 3

Circuit Elements

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Chapter 2 – Circuit Elements (1 of 9)

Review resistors:

- Ohm's Law
- KCL
- KVL
- Solving circuits

Let's do some examples of how to solve circuits containing independent sources, dependent sources and resistors.

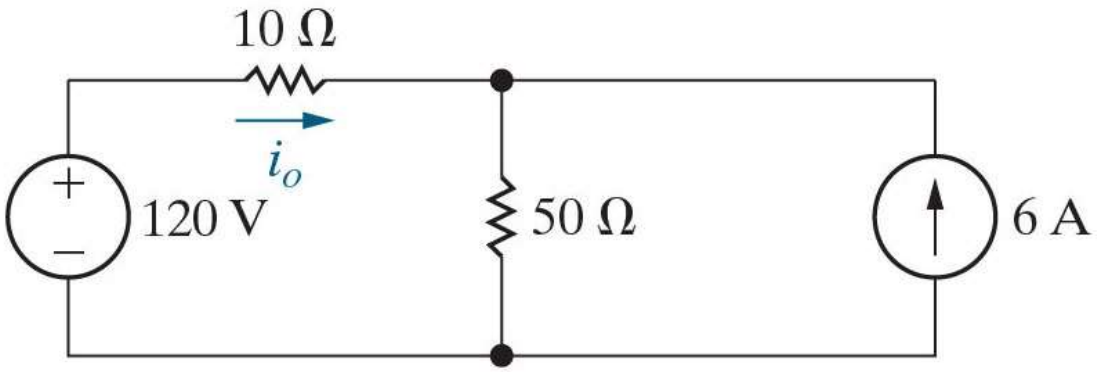
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Chapter 2 – Circuit Elements (2 of 9)

Example:

- Find the current i_o and show that the power in this circuit balances.



Begin by labeling all of the unknown voltages and currents and all of the nodes on the copy of this circuit on the next slide. Figure out how many unknowns you have. Then use Ohm's Law, KCL, and KVL to write the independent equations you need.

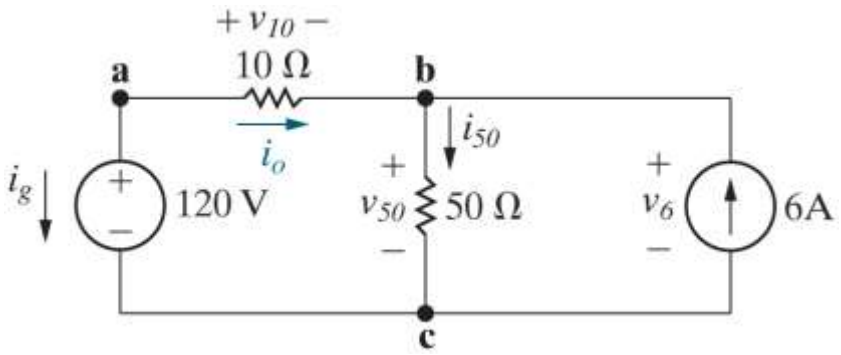
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Chapter 2 – Circuit Elements (3 of 9)

Example (continued)

- 3 unknown voltages and
- 3 unknown currents
- So we need 6 equations



Ohm's Law: $v_{50} = 50i_{50}$; $v_{10} = 10i_o$

KCL: At a: $i_g + i_o = 0$; At b: $i_o + 6 = i_{50}$

KVL: left: $-120 + v_{10} + v_{50} = 0$; right: $-v_{50} + v_6 = 0$

Solve KCL at b for i_{50}: $i_{50} = i_o + 6$

Substitute Ohm's Law equations in KVL left:

$$10i_o + 50i_{50} = 120$$

Substitute for i_{50} and solve:

$$10i_o + 50(i_o + 6) = 120 \quad \Rightarrow \quad 60i_o = -180$$

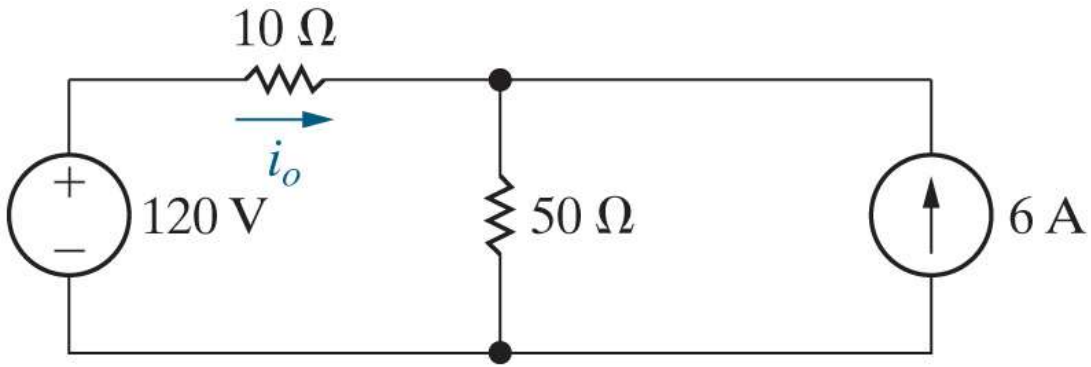
$$\Rightarrow \quad i_o = -3 \text{ A}$$

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Chapter 2 – Circuit Elements (4 of 9)

Before we start calculating the power, let's try to predict which elements will be supplying power.



✗ A. Just the resistors

✗ B. Just the sources

✗ C. One resistor and one source

✓ D. We need more information

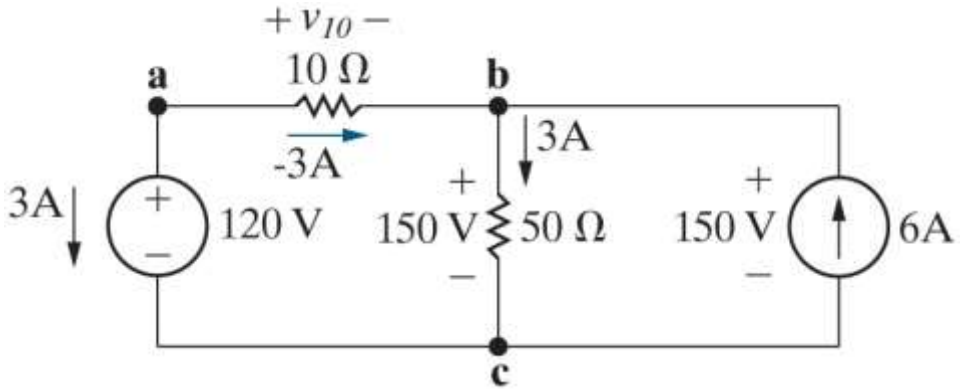
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Chapter 2 – Circuit Elements (5 of 9)

Example (continued)

- Power balance:



$$p_{120V} = (120)(3) = 360 \text{ W}$$

$$p_{10\Omega} = (10)(-3)^2 = 90 \text{ W}$$

$$p_{50\Omega} = (50)(3)^2 = 450 \text{ W}$$

$$p_{6A} = -(150)(6) = -900 \text{ W}$$

Total power is 360 + 90 + 450 − 900 = 0 (check!)

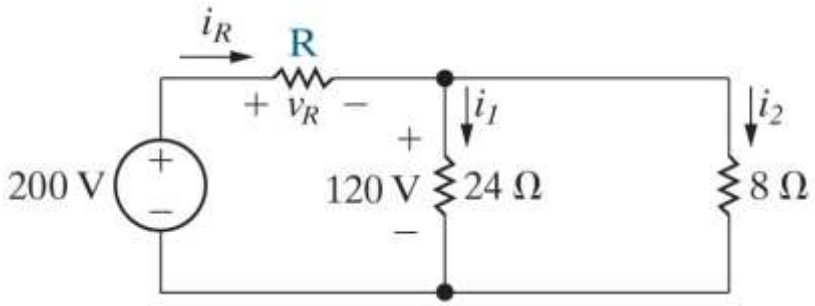
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Chapter 2 – Circuit Elements (6 of 9)

Example:

- Find *R*.



To find R we need to find the voltage drop across R and the current through R:

KVL left: $-200 + v_R + 120 = 0 \quad \Rightarrow \quad v_R = 200 - 120 = 80 \text{ V}$

KCL top: $i_R = i_1 + i_2$

KVL right: $-120 + v_2 = 0 \quad \Rightarrow \quad v_2 = 120 \text{ V}$

$$\therefore \quad i_R = \frac{120 \text{ V}}{24\Omega} + \frac{120 \text{ V}}{8\Omega} = 5 + 15 = 20 \text{ A}$$

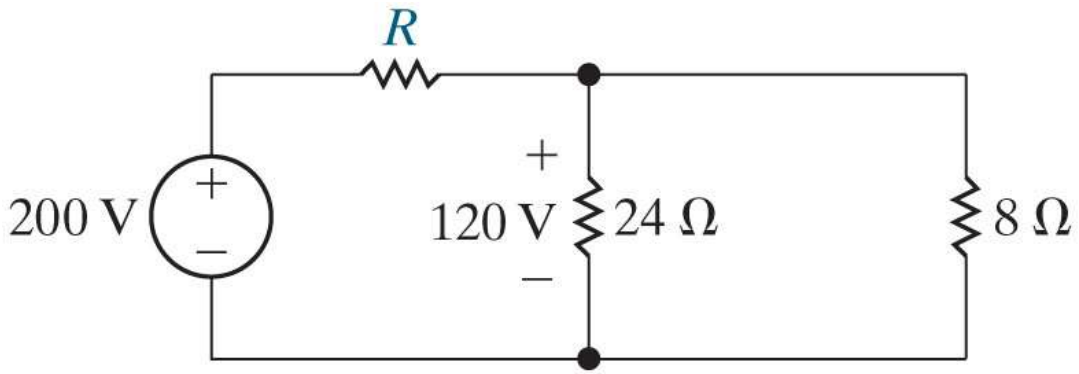
$$\Rightarrow \quad R = \frac{v_R}{i_R} = \frac{80}{20} = 4\Omega$$

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Chapter 2 – Circuit Elements (7 of 9)

Let's see if we can predict which elements will be supplying power in the circuit below.



- ✗ A. Just the resistors
- ✓ B. Just the source
- ✗ C. The source and one resistor
- ✗ D. We need more information

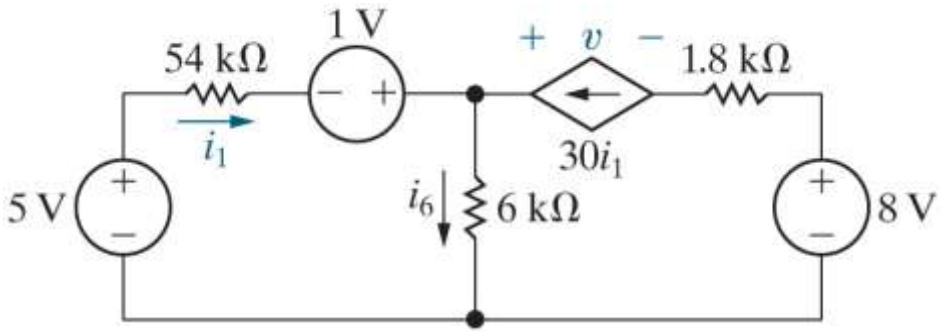
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Chapter 2 – Circuit Elements (8 of 9)

Example:

- Find i_1, v, and show that the power balances.



KVL left: $-5 + 54{,}000i_1 - 1 + 6000i_6 = 0$

KCL top: $i_1 + 30i_1 = i_6 = 31i_1$

$\Rightarrow \quad -5 + 54{,}000i_1 - 1 + 6000(31i_1) = 0$

$\Rightarrow \quad [54{,}000 + (6000)(31)]i_1 = 6 \quad \therefore \quad i_1 = 25\,\mu\text{A}$

KVL right: $-v + (6000)(31i_1) - 8 + 1800(30i_1) = 0$

$\Rightarrow \quad v = (6000)(31)(25\mu) - 8 + (1800)(30)(25\mu) = -2\ \text{V}$

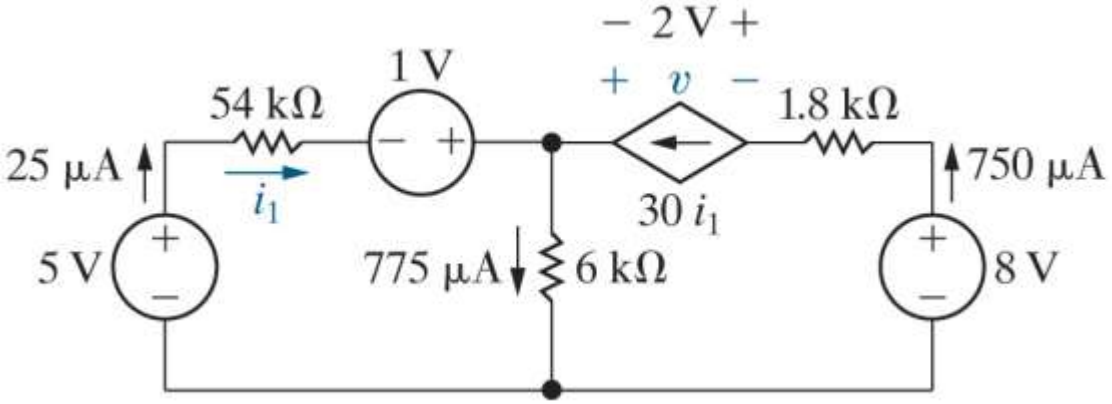
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Chapter 2 – Circuit Elements (9 of 9)

Example:(continued)

- Power balance:



$p_{5V} = -(5)(25\mu) = -120\mu\text{W}$ $\quad p_{d.s.} = (2)(750\mu) = 1500\mu\text{W}$

$p_{54k} = (55{,}000)(25\mu)^2 = 33.75\mu\text{W}$ $\quad p_{1.8k} = (1800)(750\mu)^2 = 1012.5\mu\text{W}$

$p_{1V} = -(1)(25\mu) = -25\mu\text{W}$ $\quad p_{8V} = -(8)(750\mu) = -6000\mu\text{W}$

$p_{6k} = (6000)(775\mu)^2 = 3603.75\mu\text{W}$

The sum of the power is

− 120 + 1500 + 33.75 + 1012.5 − 25 − 6000 + 3603.75 = 0!

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